



Data Science, Job Opportunities, Learning R and Data Analysis with R Applications in Economics

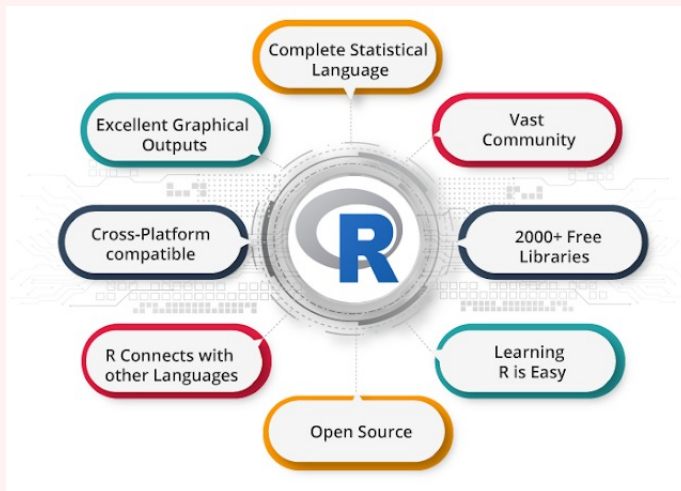
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Objectives of the Presentation

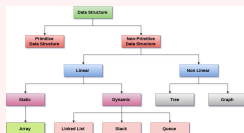
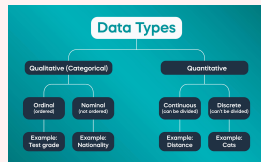
- Learn about R-studio make and analysis with data sheet.
- Making collaboration R studio with economics.
- Learning about the Job Opportunities.
- Learn about R-studio make and analysis with data sheet.
- Data security and privacy.

Significance of the Presentation



Introduction To Data Science

- ① Data is a collection of facts, like numbers, words, measurements and observations .



- ② As an economic analyst I can use data science to: 1) Deal with large set 2) Summarize data and take economic decisions 3) Represent economic models easily

Importance and Popularity of R and R-Studio and How to Master a Programming Language

- ➊ R and R studio is a free software programming language
- ➋ Open source allows them to test, experiment, and iterate more than most proprietary software.
- ➌ Open source data science tools are most popular because of it is easy to collaborate projects and modify programs in those tools.
- ➍ R and R studio is a constantly being updated and also growing programming language.
- ➎ Pick a programming language as per your demand.
- ➏ Strengthen your fundamentals.
- ➐ Proceed from basics to advanced level.
- ➑ There is no alternative to practice.
- ➒ Developing projects and solving problems.
- ➓ Sharing knowledge and code with others.

Job Opportunities for Economists in the Field of Data Science and Exploring Data Science Jobs

No.	Position	location	Company Name	Salary
1	Data Scientist	Washington, DC	Radiant.digital	\$120K- \$125K
2	Data Science Director	Newark,NJ	Prudential	\$107K- \$153K
3	Economic Data Analyst	Austin,TX	Realtor.com Careers	\$59K - \$84K
4	Senior Data Analyst	Dhaka,BD	Deriv	\$30K-\$45K
5	Financial Analyst	Bangladesh	CrossFund	\$35K-\$50K

- 1 Create a Compelling Resume and a portfolio.
- 2 Create Industry-Specific Projects.
- 3 Approach Employers and do networking.
- 4 Look For Entry-Level Data Science, Research Intern Jobs.
- 5 Consider Working Remotely.
- 6 Build Your Personal Brand.

Tools, Resources and Creating Content for Community and Their privacy and Security

- 1 Excel, SPSS, SQL, Python, R, Java are some essential tools to a career in data science.
- 2 Youtube channels: Alex the Analyst, Luke Barousse, Ken Jee, Recall by Dataiku, Tina Huang, freeCodeCamp.org, Guy in a Cube.
- 3 Analytic Blogs: KDnuggets, Dataflog, Dataeconomy, Data and information visualization, Insidebigdata.
- 4 Content creator can provide their contents to the Data science community by using Github, KDnuggets, Kaggle, LinkedIn, Medium, etc.
- 5 Data security is the concept of protecting digital data from theft, corruption, or unauthorized access.
- 6 Data privacy enables individuals to decide and limit access to the use and sharing of their personal data.

R-Studio Clouding System

- ➊ R-Studio Cloud is basically a cloud based solution system.
- ➋ R-Studio Cloud runs on a remote Server.
- ➌ R-Studio Cloud allows anyone online to access R-Studio through a web browser.
- ➍ R-Studio Cloud made it easy to share, teach and learn data science.

Designing Online Data Science Training for the Modern Age

- ➊ In modern age, we must go beyond typical e-learning method.
- ➋ The e-learning process should be Outcome-based, comprehensive, iterative and inclusive
- ➌ Modern learners must have a network for proper guideline online.
- ➍ One platform for implementing this method is R-Studio Academy.

Necessary Rules for Learning R by Yourself

Taking Time to Read Books

- Free ebooks in online
- Code examples, Exercises
- Solutions available in online
- Include advanced statistical skills and core programming of R.
- Give knowledge about web-based data visualization techniques.

Using Free Resources

- Low cost of learning
- R-Studio provide cheat sheets
- Free courses available on sites like Coursera, eDX.
- Blog post, workshop and resources offered by R groups.
- Many institution offer free tutorials, workshops and resources.

Necessary Rules for Learning R by Yourself

Adopting Good Practices and Being Consistent Joining the R Community

- Consider mindset while learning R
- Pseudocode might help at the very beginning
- One may choose learning by practically doing code
- Commenting is necessary to understand the code in the future
- A common style would be the best choice.
- prolific, with conferences, meetups, and regular online events
- Local UseR or RLadies groups, near an R group location, the R4DS Slack channel
- R bloggers a great way to stay up-to-date with R news.
- Online a great way to learn about new features, stay informed, come across tips and tricks and stay motivated.

Necessary Rules for Learning R by Yourself

Reading and Sharing One-another's Code

- If we find any R product in online, we should run the code to understand the exact benefaction.
- Sharing our R code in online will help us become a more proficient coder.

Never Box Yourself In

- Proficiency in R will help us to gain skills that can be applied to other programs.
- The determination we build while learning R can be the begining of venture of our coding journey.

Data Analysis Part-1

1. Smoke Dataset

2. **Ten Variables:** educ, cigpric, white, age, income, cigs, restaurn, lincome, agesq, lcigpric

3. Summary Statistics:

Lincome:	Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
	6.215	9.433	9.903	9.687	10.309	10.309

Cigpric:	Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
	44.00	58.14	61.05	60.30	63.18	70.13

Cigs:	Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
	0.000	0.000	0.000	8.697	20.000	80.000

ANOVA Hypothesis:

H_0 : *Cigpric we have no significant effect on lincome*

H_0 : *Cigs will have no significant effect on lincome*

H_0 : *Cigpric and Cigs will have no significant effect on lincome*

Data Analysis Part-3

- I Smallest, largest and average years of schooling are 6, 18 and 12.47 respectively
- II The number of people don't smoke and claim to smoke 20 cigarettes smoked per day are 497 and 101 respectively. 21.15% people claim to smoke 20 cigarettes a day

III

Shapiro Wilk Normality test : $w = 0.68741$, $p - value < 2.2e^{-16}$

IV

Average income = 19304.83

The average is not a good measure because there are outliers or extreme value present in the data.

Average number of cigarettes smoked per day where restaurant smoking apply if 6.6.

Data Analysis Part-4

Variable	DF	SS	MS	F-value	P-value
Cigpric	1	2.3	2.3016	4.561	0.0330
Cigs	1	1.8	1.8295	3.625	0.0573
Cigpric:Cigs	1	0.0	0.0458	0.091	0.7632
Residuals	803	405.2	0.5046	-	-

Data Analysis Part-5

Variable	P-Value	Decision
Cigpric	0.0331	Significant
Cigs	0.565	Not Significant
Cigpric:Cigs	0.7633	Not Significant

Thank You!

Questions and Answers?